

The 'sopra-tables' package

Documentation for the 'sopra-tables' package | Version v1.0.04

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1 General

1.1 Why, wherefore, whence?

This $\text{\LaTeX}$ 2_ε-package was written as part of the Sopra project in the winter semester of 2019 and summer semester of 2020, and serves as the basis for creating tables for *Team 20*. This documentation was created together with `sopra-base.cls` and the `sopra-documentation.sty` package.

The `sopra-listings` package is used to visualize the individual code snippets. The corresponding package should also be embedded in this document: .

1.2 Dependencies

This package includes the following packages:

- `tabu`^(usetabu)
- `colortbl`^(usetabu)
- `booktabs`^(notabu)
- `makecell`
- `longtable`
- `amssymb`
- `xcolor`

All of these packages should be part of any common $\text{\LaTeX}$ -distribution.

1.3 Installation

The package is not distributed as a `.dtx` file, which is why the following options are available:

- The package can be placed in the same directory as the document. In this case, the inclusion command reads:
`\usepackage{sopra-tables}`
- The package can be placed in a subdirectory, or in a directory shipped alongside the document. In this case, it is specified via the (relative) path:
`\usepackage{./My/Path/to/sopra-tables}`
- The package can be placed (via a symlink or similar) into your own `texmf` tree. For example, on Linux using `texlive`, the package can be placed here: `~/texmf/tex/latex/`. The directory can be created and then updated using `texhash ~/texmf`. The package can now be used like any other installed package:
`\usepackage{sopra-tables}`

1.4 Further Notes

In version v1.0.04 (`\thesotversion` → p. ??) there are no further notes.

1.5 Accepted Parameters

Like most packages, this package accepts arguments. For arguments with a “Counter” option, the one active by default is written first and the other in parentheses. So the following is called implicitly:

```
\usepackage[usetabu]{sopra-tables}
```

While with:

```
\usepackage[notabu]{sopra-tables}
```

we compile the document without the tabu package.

▷ usetabu (notabu)

Analogous to the notabu option of sopra-requirements.sty: by default, this package uses tabu for visual reasons to render the tables. Should the (bug) fixes contained in this package (no longer) work, or should the booktabs tables used in notabu be preferred, this option can be switched accordingly. Note: With usetabu, `\relax` is set within `env@mtabular` so that no error/horizontal lines occur when one wants to write for both designs.

▷ cpalette (nocpalette)

Offers support for <https://github.com/Eagleoutlce/color-palettes>. In this case, the table's highlighting is automatically adapted to the current palette.

▷ invert (noinvert)

Inverts the highlighting for the header row.

1.6 Colors

With usetabu, the color `MaterialHeaderColor` defines the color of the title column and `NextMaterialHeaderColor` the color in case of a page break. The colors can be changed like any other. Within sopra-base, a color adapted to the design is chosen automatically.

2 Commands and Environments

Note that the prefix `env@` only indicates the nature of an environment and is not part of the actual identifier!

2.1 The Tables

▷ `env@mtabular`{Column definition}

Sets a tabu/tabular table, where the first row automatically becomes the title row and is highlighted accordingly.

▷ `env@mltabular`[Header]{Column definition}

Sets a longtabu/longtable table, where specifying `Header` defines the header column, which is always appended in case of a page break. Should this not be desired, it suffices to not specify the `Header` and simply write the first row as the title row as usual, which is not placed at the start of every page. The restrictions known and typical for longtable apply.

2.2 General Notes and Tips

For a `longtabu`/`longtable` to be displayed correctly, a second compilation pass is usually required.

It is recommended to use the `env@table` environment together with `\caption` to embed the table sensibly. `env@wrapfigure` is also an option.

If the `sopra-requirements` package is also included, the column types `K` and `V` can be used for two-column tables.

2.3 Example

In the following, two tables are illustrated; note that `env@mltabular`^{→ p. 2} is automatically centered by its very nature.

```
1 \begin{mtabular}{lcr}
2   Hello & World & \texttt{How}'s it going? \\
3   How are & you doing & today then? \\
4   Bi & Bu & bo \\
5   Hi & Hi & Hi \\
6 \end{mtabular}
```

Gives:

Hello	World	How's it going?
How are	you doing	today then?
Hi	Hi	Hi
Bi	Bu	bo

And similarly:

```
1 \begin{mltabular}[I & am & a Header! & Epic-Row]%
2   {lcr>\raggedleft\arraybackslash m{5cm}}
3   Hello & World & \texttt{How}'s it going? & HELLO EARTH \\
4   How are & you doing & today then? & Hi \\
5   Bi & Bu & bo & You can tell I'm \\
6   Suuper & Long table & because reason & really creative \\
7   A1 & B8 & C1 & D8 \\
8   A2 & B7 & C2 & D7 \\
9   A3 & B6 & C3 & D6 \\
10  A4 & B5 & C4 & D5 \\
11  A5 & B4 & C5 & D4 \\
12  A6 & B3 & C6 & D3 \\
13  A7 & B2 & C7 & D2 \\
14  A8 & B1 & C8 & D1 \\
15 \end{mltabular}
```

Gives *Note: the extra spacing here is artificial and serves to illustrate a page break:*

I	am	a Header!	Epic-Row
Hello	World	How's it going?	HELLO EARTH
How are	you doing	today then?	Hi
Bi	Bu	bo	You can tell I'm
Suuper	Long table	because reason	really creative
A1	B8	C1	D8
A2	B7	C2	D7
A3	B6	C3	D6
A4	B5	C4	D5
A5	B4	C5	D4
A6	B3	C6	D3
A7	B2	C7	D2
A8	B1	C8	D1